

Weibull distribution

A continuous random variable x , is said to follow a weibull distribution with parameter $\alpha, \beta > 0$, if it has the following p.d.f

$$f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \alpha, \beta > 0 \\ 0, & \text{Otherwise} \end{cases}$$

here $X \sim W(\alpha, \beta)$

* when $\beta=1$, it converts to exponential distⁿ

$$f(x) = \begin{cases} \alpha e^{-\alpha x}, & x > 0 \\ 0, & o.w \end{cases}$$

exp distⁿ with parameter α

Mean of Weibull distribution:-

$$\mu_1 = \text{Mean of } X = \alpha^{-1/\beta} \left[1 + \frac{1}{\beta} \right] \quad \left[\mu_2' = e^{-\frac{2}{\beta}} \left[1 + \frac{2}{\beta} \right] \right]$$

Variance of Weibull distribution:-

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_2 = \text{Variance of } X = \alpha^{-\frac{2}{\beta}} \left[\left(1 + \frac{2}{\beta} \right) - \left(1 + \frac{1}{\beta} \right)^2 \right]$$

Q-1 The length of life X , in hours of an item in a machine shop has a Weibull distribution with $\alpha = .01$ and $\beta = 2$. What is the probability that it fails before eight hours of usage?

Solⁿ: $X \sim$ length of life
 $X \sim W(\alpha = .01, \beta = 2)$

The P.d.f of X

$$f(x) = \frac{1}{100} * 2 * x^{2-1} e^{-\frac{x^2}{100}}, \quad x > 0$$

$$f(x) = \frac{1}{50} x e^{-\frac{x^2}{100}}, \quad x > 0$$

To find, probability that items fails before eight hour of usage

$$\text{ie } P(X < 8) = \int_0^8 \frac{1}{50} x e^{-\frac{x^2}{100}} dx$$

$$= \frac{1}{50} \quad \text{put } x^2 = t$$

$$2x dx = dt$$

$$\text{at } x=0, \quad t=0$$

$$\text{at } x=8 \quad t=64$$

$$= \int_0^{64} \frac{e^{-t}}{50} \frac{dt}{2}$$

$$= \frac{1}{100} \int_0^{64} e^{-t} dt = \frac{1}{100} - (e^{-t})_0^{64}$$

$$= \frac{1}{100} (1 - e^{-64}) \quad \text{Ans}$$

Q: Each of the six tubes of a radio set has a life length (in years), which may be considered as random variable that follows weibull distribution, with parameter $\alpha=25$, $\beta=2$. If these tubes function independently on one another, what is the probability that no tube will have to be replaced during first two month of service.

Soln:- $f(x) = \alpha \beta x^{\beta-1} e^{-\alpha x^{\beta}}$, $x > 0$, $\beta > 0$

given $\alpha=25$, $\beta=2$

$$f(x) = 50x e^{-25x^2}, \quad x > 0$$

X is a random variable, denotes life of tubes in years.

Probability that a tube not to be replaced in first two month. i.e. for that $X > \frac{2}{12}$

$$\text{i.e. } P(X > 1/6) = \int_{1/6}^{\infty} 50x e^{-25x^2} dx$$

put $25x^2 = t$
 $50x dx = dt$

when $x = 1/6$, $t = 25/36$

$x = \infty$, $t \rightarrow \infty$

$$\begin{aligned} &= \int_{25/36}^{\infty} e^{-t} dt = - (e^{-t})_{25/36}^{\infty} \\ &= - (0 - e^{-25/36}) = e^{-25/36} \\ &= 0.4993 \end{aligned}$$

Prob. that all tubes are not replaced during the first two months

$$= (0.4993)^6 = 0.0155 \text{ Ans}$$

Q:- The life of a component measured in hours follows Weibull distⁿ with parameters $\alpha=0.3$, $\beta=0.5$. Find mean life time of the component.

Solⁿ $\alpha=0.3$, $\beta=0.5$

$$\text{Mean} = \alpha^{-1/\beta} \left| 1 + \frac{1}{\beta} \right|$$

$$= (0.2)^{-1/0.5} \left| 1 + \frac{1}{0.5} \right|$$

$$\boxed{\text{Mean} = 50 \text{ hours}}$$

Q:- If the life X (in years) of a certain type of car has a Weibull distribution with the parameter $\beta=2$. Find the value of the parameter α , given that prob. that the life of the car exceeds 5 years is $e^{-0.25}$. Find the value of α . Find mean and variance of X .

Solⁿ P.d.f of X is given by

$$f(x) = \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, \quad x > 0$$

given $\beta = 2$

$$f(x) = 2\alpha x e^{-\alpha x^2}$$

To, find α , such that $P(X > 5) = e^{-0.25}$

$$\text{ie } \int_5^{\infty} f(x) dx = e^{-0.25}$$

$$\int_5^{\infty} 2\alpha x e^{-\alpha x^2} dx = e^{-0.25}$$

$$\text{put } x^2 = t$$

$$2x dx = dt$$

$$\int_{25}^{\infty} \alpha e^{-\alpha t} dt = e^{-0.25} \Rightarrow (-e^{-\alpha t})_5^{\infty} = e^{-0.25}$$

$$-[0 - e^{-25\alpha}] = e^{-0.25}$$

$$\text{ie } e^{-25\alpha} = e^{-0.25}$$

$$25\alpha = 0.25$$

$$\alpha = \frac{0.25}{2500}$$

$$\boxed{\alpha = \frac{1}{100}}$$

$$\text{Mean of } X = \alpha^{-1/\beta} \sqrt{1 + \frac{1}{\beta}} = \left(\frac{1}{100}\right)^{-1/2} \sqrt{1 + \frac{1}{2}}$$

$$= \sqrt{100} \sqrt{\frac{3}{2}}$$

$$= 10 \cdot \frac{1}{2} \sqrt{\pi}$$

$$\boxed{\text{Mem of } X = 5\sqrt{\pi}}$$

$$\text{variance of } X = \alpha^{-2/\beta} \left[\sqrt{1 + \frac{2}{\beta}} - \left(\sqrt{1 + \frac{1}{\beta}} \right)^2 \right]$$

$$= \left(\frac{1}{100} \right)^{-1} \left(\sqrt{2} - \left(\sqrt{\frac{3}{2}} \right)^2 \right)$$

$$= 10 \times \left(1 - \left(\frac{1}{2} \sqrt{\pi} \right)^2 \right)$$

$$= 10 \times \left(1 - \frac{\pi}{4} \right)$$

$$\boxed{\text{Variance of } X = 10 - \frac{5\pi}{2}}$$

Q Suppose that the service life, in years of a hearing aid battery is a random variable having a Weibull distribution with $\alpha = \frac{1}{2}$ and $\beta = 2$

(a) How long can such battery be expected to last.

(b) what is the prob. that such a battery will be operating after 2 years?

soln x is a random variable, denotes the service life (in year) of hearing aid battery given

$$x \sim W\left(\frac{1}{2}, 2\right) \quad \alpha = \frac{1}{2}, \quad \beta = 2$$

P.d.f is given by

$$f(x) = \alpha \beta x^{\beta-1} e^{-\alpha x^2}$$

$$f(x) = \frac{1}{2} \times 2 \times x^{2-1} e^{-\frac{1}{2}x^2}, \quad x > 0$$

$$f(x) = x e^{-x^2/2}, \quad x > 0$$

(a) Expected value of battery is

$$E(x) = \alpha^{-1/\beta} \sqrt{1 + \frac{1}{\beta}}$$

$$E(x) = \left(\frac{1}{2}\right)^{-1/2} \sqrt{1 + \frac{1}{2}}$$

$$E(x) = (2)^{1/2} \sqrt{\frac{3}{2}}$$

$$E(x) = \sqrt{2} \cdot \frac{1}{2} \sqrt{3}$$

$$\boxed{E(x) = \sqrt{\frac{3}{2}}}$$

$$(b) P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - \int_0^2 x e^{-x^2/2} dx$$

$$= 1 - \int_0^2 x^2/2 = t \quad x dx = dt$$

$$= 1 - \int_0^2 e^{-t} dt$$

$$= 1 + \left(e^{-t}\right)_0^2 = 1 + e^{-2} - 1$$

$$P(X > 2) = e^{-2} = 0.1353$$